Debs Maia Silva, Ph.D.

Data Science & Machine Learning Engineer Manager
Capital One | Leader in Machine Learning and AI Systems
linkedin.com/in/dr-debs/

missdmaia@gmail.com
+1-765-775-9061
dmaiasil.github.io

Machine Learning Engineer and Ph.D. researcher specializing in multimodal AI, distributed inference, and agentic systems. Proven track record of architecting scalable deep learning pipelines, optimizing LLM inference, and delivering high-stakes enterprise projects under extreme deadlines. Adept at leveraging rigorous data-driven experimentation to solve complex, open-ended infrastructure problems and ship user-friendly AI products from zero to production.

Work Experience

Capital One — McLean, VA

- *Manager Data Science (Pod and Tech Lead)* — *07/2025 to Present*
  Principal Associate Data Science — *06/2021 to 07/2025*
  - **Data Science/ML Tech Lead**: Drive the analytical and data science vision for a high performing team of 5 Data Scientists. Partner with product, tech, and design to translate ambiguous requests into model insights, shaping product strategy and technical decisions. Team presenter for Executive leadership. Responsible for mentorship, performance reviews, and hiring decisions.
  - **Multimodal Document Information Extraction**: Architected and deployed an end-to-end multimodal pipeline to extract structured data from highly variable inbound technical documents. Engineered a dynamic routing system using NLP techniques for upfront quality control, deploying local instances of Nemotron-parse-v1.2 for image and text extraction. Orchestrated Llama-4 for complex reasoning and inference on extracted text, with Gemma-4 acting as an automated judgment model for continuous quality evaluation. Scaled processing from manual reviews to millions of automated tests.
  - **Scalable AI Systems and Architecture**: Technical Lead for a GenAI platform translating natural text into executable code in SQL and Python. Designed robust architectures to balance rapid iteration with reliable, distributed system deployments for non-technical users. Development evaluation pipeline and benchmarks for model update and usage.
  - **Agentic Systems and Business Intelligence**: Tech Lead for multi agentic Chatbot development. Built robust metric frameworks to evaluate coherence, latency, and value delivery of AI features. Championed the use of AI tools to accelerate data visualization, triage, and iteration for users.
  - **Dynamic RAG Pipeline and Search Architecture:**: Built and deployed Capital One's first production RAG application for Compliance & Ethics. Overcame latency and relevance degradation from high-frequency database updates by engineering a custom reranker with time-decay weighting and strict time-window filters to prioritize temporal relevance over pure semantic similarity. Designed architecture to evaluate LLM-wiki structures, reducing vector update overhead.
  - **Applied NLP**: Developed Deep Learning models to extract business signals from massive transcript datasets. Improved accuracy by 40 percent.
  - **ML Modeling**: Developed the full ML life-cycle for debit fraud predictive model. Conducted feature drift SQL tests and feature engineering designs. Replaced vendor model for higher ROI and in-house explainability.
- **University of San Francisco** — San Francisco, CA
  Advisor for the Strategic AI Program — *01/2025 to 08/2026*
  - **SME Artificial Intelligence**: Advised on the development of interdisciplinary AI curricula, ensuring alignment with emerging industry trends and ethical standards.
- **Purdue University** — West Lafayette, IN
  Graduate Research Assistant | Ph.D. Researcher — *08/2017 to 05/2021*
  - **Quantitative Research and Modeling**: Developed Bayesian Additive Regression Trees and ensemble models for high-dimensional datasets, translating messy data into clear metric frameworks. Research published in **Nature Communications** and featured in global scientific media.

Education

- **Purdue University** — West Lafayette, IN
  Ph.D. Purdue College of Engineering (Focus: ML, Statistics, and Data Science)
- **Campinas University (UNICAMP)** — Campinas, Brazil
  B.S. Civil Engineering; Minor in Computer Engineering

Technical and Leadership Skills

- **Leadership:** Executive Narratives, Product Analytics, KPI Definition, Technical Hiring, Cross Team Alignment
- **AI and Machine Learning:** Multimodal LLMs, Computer Vision (Object Detection/Segmentation), Generative AI, RAG Architectures, Agentic Systems, Post-Training Evaluation; Pytorch, Langchain, Tensorflow.
- **Data Tools:** Python, SQL, Agentic Context Engineering, Claude Code, Advanced Feature Engineering, Spark, Hadoop.

Publications (Selected)

- **D. Maia Silva** et al. "The role of humidity in electricity demand," *Nature Communications*, 2020.
- **D. Maia Silva** et al. "The Goldilocks Zone in Cooling Demand," *AGU Earth's Future*, 2022.

Leadership and Presentations

- **Georgetown University (2026)**: The Reality of AI's Lifecycle
- **MAC - Modeling and Analytics Conference Capital One (2023 to 2025)**: Panelist on Metric Frameworks for Generative AI Deployment and Benchmarking.
- **SECON - Tech Conference Capital One (2025)**: Building Trust and Value in Interactive AI Systems.
- **SHPE and RTC - National Convention Rewriting the Code**: Speaker Series and Mentorship roles for engineering talent.

Honors and Awards

- Capital One **Patent**: Compliance Data and Analytics Hub Platform *2025*
- Capital One **Patent**: OneScan Multimodal Document Processor *2026*
- **Outstanding Research Award** Purdue College of Engineering *2020*